

INTERNAL — TEAM REFERENCE

Quant Desk

Quantitative Trading Research Platform — Complete User Guide

Stack	Version	Date
FastAPI · Next.js 14 · DuckDB · yfinance	1.0	June 2026

This guide covers every screen in the platform — what each formula computes, how to read the results, and how to combine the tools for a full research workflow.



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1 · Platform Overview

Quant Desk is a full-stack quantitative research platform designed for systematic, factor-driven stock selection, portfolio construction, and risk analysis. It combines academic factor-investing research with practical technical analysis, all served through a fast browser-based interface.

UNIVERSES COVERED

~12,000

US-listed tickers across NYSE, NASDAQ, and exchanges

FACTOR MODELS

13 Factors

Price-based + fundamental + macro

ANALYSIS MODULES

7 Screens

Overview, Screener, Factor Explorer, Technical, Backtest, Portfolio, Risk

Architecture at a glance

Backend: FastAPI (Python) running on `localhost:8000` — fetches market data from Yahoo Finance, caches in DuckDB, computes all analytics server-side.

Frontend: Next.js 14 App Router on `localhost:3000` — React UI with interactive charts (Lightweight Charts, Recharts).

Core Design Philosophy

- **Factor-first:** Every stock is scored on multiple dimensions (momentum, quality, value, etc.) that have documented predictive power in academic literature.
- **Cross-sectional z-scores:** All factor scores are standardized to mean 0 / std 1 within the chosen universe, so you compare stocks *relative to their peers*, not on raw numbers.
- **Backtested evidence:** The Factor Explorer lets you verify whether a factor has historically added value ($IC > 0$) before trusting it in live screening.
- **Integrated workflow:** Screen → Explore factor quality → Backtest → Optimize → Risk-check — all in one platform without leaving the browser.

2 · Data Sources & Caching

Source	Data Provided	Refresh Policy
Yahoo Finance (yfinance)	Daily OHLCV prices, fundamental ratios (P/E, P/B, ROE...), analyst ratings, earnings calendar	Delta fetch on demand; background refresh every 15 min during market hours (9 AM–4:30 PM ET), every 1 hour after-hours
Kenneth French Data Library	Fama-French 3-factor daily returns (Mkt-RF, SMB, HML, RF)	Cached as <code>ff3_daily.csv.gz</code> ; refreshed every 5 days
NASDAQ Trader FTP	Full US-listed universe (~12,000 securities), exchange, ETF flag	Fetches on first use, stored in DuckDB <code>us_universe</code> table
DuckDB (local cache)	All fetched prices (<code>prices</code> table), fundamentals (<code>fundamentals</code> , 24-hr TTL), constituents	Read-through cache; stale data is refreshed transparently

First-run latency

When analyzing a large universe for the first time (e.g., All US ~12,000 tickers), the platform queues price downloads in the background. Factor scores become available progressively as each batch is cached. Subsequent loads are fast (milliseconds from DuckDB).

3 • Module Reference

3.1 Market Overview

localhost:3000 / Home page

Daily health check — where is the broad market and which sectors are leading?

What's on screen

Panel	What it shows	How to use it
Index Cards	SPY, QQQ, IWM, VIX · 1D / WTD / MTD / YTD / 1Y returns	Quick pulse check — is the sell-off broad or narrow? IWM lagging SPY = risk-off / small-cap weakness
Candlestick Chart	1-year daily OHLCV for the selected index	Click an index card to switch. Identify trend, support/resistance, and recent price action structure
Market Breadth Panel	% of S&P 500 stocks above their 50-day and 200-day moving averages	>70% above 50 MA = healthy bull trend. <30% = oversold, possible bounce. Divergence from index (index up, breadth down) = distribution warning
Sector Heatmap	11 GICS sectors colour-coded by return · 1D / 1W / 1M / 3M / YTD toggle	Identify sector rotation: which sectors are accumulating money vs being distributed. Green + high = institutional flow; red + persistent = avoid longs there

Tip: Breadth divergence

If SPY makes a new high but fewer than 60% of stocks are above their 50 MA, the rally is being carried by a small group of mega-caps. This is a risk signal — consider reducing position size or tightening stops on new longs.

3.2 Universe Screener

localhost:3000/screener

Rank every stock in a universe across all factor dimensions simultaneously.

Universe Options

S&P 500 (~503 stocks)

All US Listed (~12,000)

Themes (9 baskets)

Universe	Best for	Notes
S&P 500	Institutional-grade screening; liquid names only	Scores refresh every 5 min. Use for most daily workflows
All US Listed	Catching small/micro-cap opportunities	First load is slow (price downloads queued). Use Exchange + Asset Type filters to narrow
Themes	Relative strength within a sector basket	Z-scores are computed within the theme, not vs S&P 500. Best for finding the strongest stock inside e.g. "AI Infrastructure"

Factor Score Columns

Column	Factor	High score means...
Mom12-1	Momentum 12-1M	Strong 1-year price performance (excluding last month)
Mom6-1	Momentum 6-1M	Strong 6-month price performance
LowVol	Low Volatility	Lower-than-average realized volatility — defensive/stable
Liq	Liquidity	Higher average daily dollar volume — easier to trade
Macro	Macro Regime (Low Beta)	Low market beta — defensive in risk-off environments
Value	Value	Cheap on P/E, P/B, P/S multiples
Quality	Quality	High ROE, strong margins, low debt
Prof	Profitability	High operating margins and ROA
ERev	Earnings Revisions	Positive earnings and revenue growth momentum
Sent	Sentiment	Analyst buy consensus, high price-target upside
Composite	Equal-weight average	Best overall multi-factor score — start here

How to read the Z-scores

+1.0 = one standard deviation above average for this universe. A composite Z-score above +1.5 puts a stock in roughly the top 7% of its peer group. Scores below −1.0 are bottom quartile. Focus on stocks with a Composite Z > +0.8 across multiple individual factors (not just one outlier factor).

Buttons

- **Fetch Fundamentals** — loads P/B, P/E, P/S, ROE, margins, analyst ratings for the top 100 visible tickers. Enables the Value, Quality, Profitability, Sentiment, and Earnings Revisions columns.
- **Fetch Prices** — queues background downloads for any tickers missing from the local cache.

3.3 Factor Explorer

localhost:3000/factors

Validate whether a factor has historically predicted returns before trusting it in screening.

Two types of factors

- Price-based (solid border) — Full IC history available
- Fundamental (dashed border) — Latest scores only

Price-based factors (Momentum, Volatility, Liquidity, Macro) can be back-computed historically from price data — so you can see how predictive they were in each past month. Fundamental factors depend on quarterly filings, which limits historical IC calculation.

Key Statistics

Metric	Formula	Good threshold
Mean IC	Average monthly Spearman rank correlation between factor scores and 21-day forward returns	0.02 – 0.08 (2–8%)
ICIR	Mean IC ÷ Std(IC) — signal-to-noise ratio	> 0.5 indicates a reliable signal
% Positive IC	Fraction of months where IC > 0	> 55% (factor works more months than not)

Charts

- **IC Time Series** — monthly IC bars + 3-month rolling average. Look for periods where IC collapsed — those are regime shifts where the factor stopped working.

- **Quintile Returns** — stocks sorted into 5 buckets by factor score at each month-end, then tracked for 21 days. A monotonic $Q1 > Q2 > Q3 > Q4 > Q5$ spread confirms the factor has real predictive power.

Tip: Regime-conditional factors

Momentum tends to have positive IC in trending markets and negative IC (reversal) after sharp sell-offs. Low Vol tends to outperform in high-VIX environments. Use the IC time series to understand when a factor works — not just whether it works on average.

3.4 Technical Signals Screener

localhost:3000/intraday

Multi-indicator technical screen — find stocks at actionable setups with timing alignment.

Column Glossary

Column	Formula Summary	Reading
RSI	14-day Relative Strength Index	<30 oversold (blue), >70 overbought (red)
MACD Hist	MACD Line – Signal Line (12/26/9)	Positive & rising = building momentum
BB%B	$(\text{Price} - \text{Lower Band}) \div (\text{Upper} - \text{Lower}) \cdot 20d/2\sigma$	>1 extended high; <0 extended low
MA50%	$(\text{Price} \div 50\text{-day MA}) - 1$	Positive = above trend. Negative = below
MA200%	$(\text{Price} \div 200\text{-day MA}) - 1$	Long-term trend orientation
RS 20d	Stock 20d return – SPY 20d return	Positive = outperforming market recently
RS 5d	Stock 5d return – SPY 5d return	Short-term relative strength
Volx	Today's volume ÷ 20-day avg volume	>2× highlighted in amber — unusual activity
ATR%	14-day ATR ÷ Price	Volatility per share, normalised to price level
Pivot	Nearest monthly pivot level (PP, R1–R3, S1–S3)	Key inflection zone

Column	Formula Summary	Reading
Pvt Dist	$(\text{Price} - \text{Pivot Level}) \div \text{Price}$	How far from the nearest pivot as a % of price
Score	Composite z-score of all technical signals	Higher = more multi-indicator alignment

Setup Classifications

Setup	Criteria	Trade idea
Early Breakout	RS strong, near pivot resistance, volume $> 1.5\times$	Enter on breakout above pivot with tight stop below it
Volatility Squeeze	BB width compressed, drying volume, Stage 1/2	Anticipate directional move; wait for volume + range expansion
Momentum Continuation	Above both MAs, MACD+, RSI 50–75, Stage 2	Add on pullbacks to 50 MA while trend intact
Institutional Accumulation	High volume, flat intra-day range, above 50 MA	Potential large buyer absorbing supply; follow the volume
Mean Reversion Bounce	RSI < 35 , near support, Stage 1/2	Counter-trend long; tighter position sizing, quicker exits
Failed Breakdown Reversal	Near 52W low but RS recovering, volume returning	High-risk / high-reward reversal; confirmation required

Near-Pivot Filter

Toggle **Near Pivot** to show only stocks within your chosen threshold (1–3%) of a monthly pivot level. These are the highest-priority setups — the stock is approaching a zone where big players have historically reacted.

Multi-Timeframe Alignment Score

Scores alignment across weekly (40%), daily (35%), and short-term (25%) timeframes:

Weekly: 10-week MA position + MA slope + RSI(9)

Daily: Above 50 MA + Above 200 MA + RS vs SPY 20d

Short-term: Above 5 MA + 5-day up + confirming volume

A score near 100 means all three timeframes point in the same direction — the highest-conviction setups.

3.5 Strategy Backtester

localhost:3000/backtest

Test whether a factor-based long-only strategy would have outperformed the S&P 500 historically.

Strategy mechanics

- **Universe:** S&P 500 constituents
- **Selection:** Top N stocks by chosen factor score at each rebalance date
- **Weighting:** Equal weight within selected stocks
- **Rebalance:** Monthly (end-of-month)
- **Benchmark:** SPY total return
- **Transaction cost:** User-specified round-trip bps applied on each rebalance turnover

Input Parameters

Input	Typical range	Notes
Factor	Any from the factor registry	Try Momentum 12-1M as a baseline; it's one of the most robust academically
Top N	10–50 stocks	Lower N = more concentrated / higher tracking error. 20–30 balances capacity and signal
Transaction Cost	5–20 bps	10 bps is realistic for institutional. Use 20–25 bps for retail or illiquid names
Start Date	As far back as data allows	Use full history for statistical significance; watch for survivorship bias near ends

Output Metrics

TOTAL RETURN \$-growth Terminal value of \$1 invested	CAGR Annualised % Compounded annual growth rate	SHARPE RATIO Return / Vol Risk-adjusted return (higher = better)
SORTINO RATIO Return / Downside Only penalises bad volatility	MAX DRAWDOWN Peak-to-trough % Worst loss from any peak	INFORMATION RATIO Active return / TE

Consistency of outperformance vs SPY

Tip: What to look for

A Sharpe above 0.7 and positive alpha with IR > 0.5 suggests a genuine edge. Check the Rolling 12M Sharpe chart — if it's consistently positive across different market cycles (not just a single lucky bull run), the factor has more robust predictive power.

3.6 Portfolio Optimizer

localhost:3000/portfolio

Given a set of tickers, find the scientifically optimal allocation using Modern Portfolio Theory.

How to use

1. Paste comma-separated tickers from your screener shortlist (e.g. `AAPL, MSFT, NVDA, JPM, XOM`)
2. Set a maximum per-position weight (default 25%) to enforce diversification
3. Click **Optimize** — four methods are computed in parallel

Optimization Methods Compared

Method	Objective	Best for
Equal Weight	1/N allocation	Baseline comparison; surprisingly hard to beat out-of-sample
Max Sharpe	Maximise (Expected Return) ÷ Volatility	When you have high conviction in the return estimates
Min Volatility	Minimise portfolio standard deviation	Risk-averse portfolios; tends to concentrate in low-correlation assets
HRP (Risk Parity)	Equalise risk contribution via hierarchical clustering	Robust to estimation error; performs well out-of-sample

Output panels

- **Efficient Frontier Scatter** — every possible portfolio plotted by risk vs return; the curve shows the maximum return achievable at each volatility level. Each method's portfolio is marked on the frontier.

- **Weight Table** — side-by-side allocation for all methods. Large differences between Max Sharpe and Equal Weight signal the optimizer is making a strong tilt — verify the underlying assumptions.
- **Correlation Heatmap** — dark red = highly correlated (redundant). If two holdings are >0.8 correlated, consider replacing one for a name that adds diversification.

Warning: Mean-variance sensitivity

Max Sharpe allocations can be unstable — small changes in input data produce very different weights. The HRP method is more robust for live portfolios. Always sanity-check the Efficient Frontier and look for corner solutions (single-stock concentration) before using the weights.

3.7 Risk Dashboard

localhost:3000/risk

Decompose what drives your portfolio's returns and measure downside risk.

How to use

Enter your current holdings in the text box as `TICKER: weight` pairs, one per line. Weights should sum to 1.0.

```
AAPL: 0.25
MSFT: 0.20
NVDA: 0.15
JPM: 0.10
XOM: 0.10
CASH: 0.20
```

Panels

Panel	What to look for
Portfolio Stats	Overall annualized return, volatility, Sharpe ratio over the look-back period
VaR / CVaR	At 95% confidence: on a bad day, you can expect to lose no more than this amount. CVaR is the average loss on the days that exceed VaR — the true tail risk. Compare historical vs parametric: a large gap means your return distribution has fat tails
Fama-French Attribution	Which factors explain your returns? High β_{Mkt} = you're mostly taking market risk. Positive α = you're generating alpha beyond factor exposure. See formulas section for

Panel	What to look for
	details
Rolling Beta	How your portfolio sensitivity to the market changes over time. A rising beta during sell-offs is dangerous
Sector Exposure	Breakdown by GICS sector — check for unintended sector concentration
Concentration (HHI)	Herfindahl Index approaching 1.0 = single-stock portfolio. Effective N shows the equivalent number of equal-weight positions. Aim for Effective N > 8 for reasonable diversification
Correlation Heatmap	Identify pairs that cluster together — these will all move against you simultaneously in a drawdown

4 • Fundamental Factor Formulas

All factors are computed cross-sectionally within the chosen universe and then z-scored (mean=0, std=1). Extreme values are winsorized at ± 3 standard deviations to prevent outliers from dominating.

Z-score normalisation:

$$Z(x) = (x - \text{mean}(x)) / \text{std}(x)$$

Then clip: $Z = \max(-3, \min(3, Z))$

Composite Score:

$$\text{Composite} = \text{mean}(\text{all available Z-scores})$$

Momentum 12-1M

Captures the tendency of recent winners to keep outperforming. Skips the most recent month to avoid short-term reversal effects.

$$\text{Momentum}_{12_1} = \text{Price}[\text{today} - 21 \text{ days}] / \text{Price}[\text{today} - 252 \text{ days}] - 1$$

Lookback: 252 trading days (~12 months)

Skip: 21 trading days (~1 month)

Momentum 6-1M

$$\text{Momentum}_{6_1} = \text{Price}[\text{today} - 21 \text{ days}] / \text{Price}[\text{today} - 126 \text{ days}] - 1$$

Lookback: 126 trading days (~6 months)

Skip: 21 trading days (~1 month)

Low Volatility

Historically, low-volatility stocks have produced surprisingly competitive risk-adjusted returns — the "Low Vol anomaly".

$$\text{DailyLogReturns} = \log(\text{Price}[t] / \text{Price}[t-1])$$

$$\text{AnnualisedVol} = \text{std}(\text{DailyLogReturns}, 252) \times \sqrt{252}$$

$$\text{LowVol_Score} = -\text{AnnualisedVol} \quad \leftarrow \text{inverted: lower vol = higher score}$$

Liquidity

```
DollarVolume[t] = Close[t] × Volume[t]
AvgDolVol      = mean(DollarVolume, 63 days)  ← ~3 months

Liquidity_Score = log(1 + AvgDolVol)
```

Macro Regime (Low Beta)

```
Beta = Cov(StockReturns, SPYReturns) / Var(SPYReturns)
      ← estimated over 252-day rolling window

MacroRegime_Score = -Beta          ← inverted: lower beta = higher score
# Rationale: In risk-off regimes (high VIX), low-beta stocks are more defensive
```

Value

```
Value_Score = mean(
  -Z(Price / Book),      ← inverted: lower P/B is cheaper
  -Z(Trailing P/E),      ← inverted
  -Z(Price / Sales)      ← inverted
)
```

Quality

```
Quality_Score = mean(
  Z(Return on Equity),
  Z(Gross Margins),
  -Z(Debt / Equity)      ← inverted: lower leverage = higher quality
)
```

Profitability

```
Profitability_Score = mean(
  Z(Operating Margins),
  Z(Return on Assets)
)
```

Earnings Revisions

```
EarningsRevisions_Score = mean(
  Z(Earnings Growth),
  Z(Revenue Growth)
)
```

Sentiment

```
Sentiment_Score = mean(
  Z(6 - Analyst Recommendation Mean), ← analyst ratings: 1=Strong Buy, 5=Sell.
  Inverted so Buy = higher
  Z(Target Price / Current Price - 1) ← price-target upside
)
```

Size

```
Size_Score = Z(log(Market Cap))          ← log to normalise skew
```

Information Coefficient (Factor Validation)

```
IC[month t] = Spearman_Rank_Correlation(
  FactorScores[end of month t],
  ForwardReturns[next 21 trading days]
)

Mean IC = average(IC[all months])
ICIR     = Mean IC / std(IC)          ← signal-to-noise; target > 0.5
%Pos IC  = count(IC > 0) / total months ← target > 55%
```

5 • Technical Indicator Formulas

RSI — Relative Strength Index (14-day)

```
Gains  = mean(positive daily changes over 14 days)
Losses = mean(|negative daily changes| over 14 days)
RS      = Gains / Losses
RSI     = 100 - 100 / (1 + RS)
```

Interpretation:

```
RSI > 70 → Overbought (highlighted red)
RSI < 30 → Oversold   (highlighted blue)
RSI 40-60 → Neutral trend zone
```

MACD — Moving Average Convergence/Divergence (12/26/9)

```
MACD_Line   = EMA(price, span=12) - EMA(price, span=26)
Signal_Line  = EMA(MACD_Line, span=9)
MACD_Hist   = MACD_Line - Signal_Line
```

Interpretation:

```
Histogram > 0 and rising → Bullish momentum building
Histogram > 0 and falling → Momentum slowing (potential sell signal)
Histogram < 0             → Bearish pressure
```

Bollinger Bands %B (20-day, 2σ)

```
Mid   = SMA(price, 20)
Std   = rolling_std(price, 20)
Upper = Mid + 2 × Std
Lower = Mid - 2 × Std
```

```
%B = (Price - Lower) / (Upper - Lower)
```

Interpretation:

```
%B > 1 → Price above upper band (extended)
```

$\%B = 0.5 \rightarrow$ Price at midpoint (neutral)
 $\%B < 0 \rightarrow$ Price below lower band (extended/oversold)

Moving Average Distance

$MA50_Distance = (Price / SMA(price, 50)) - 1$
 $MA200_Distance = (Price / SMA(price, 200)) - 1$

Positive $\rightarrow$ price above moving average (bullish)
 Negative $\rightarrow$ price below moving average (bearish)

ATR — Average True Range (14-day)

$TrueRange[t] = \max(\begin{aligned} &High[t] - Low[t], \\ &|High[t] - Close[t-1]|, \\ &|Low[t] - Close[t-1]| \end{aligned})$
 $ATR = SMA(TrueRange, 14)$
 $ATR\% = ATR / Price \quad \leftarrow \text{normalised for comparing across price levels}$

Volume Surge

$VolSurge = Volume[today] / SMA(Volume, 20)$
 $> 2\times \rightarrow$ unusual institutional activity (amber highlight)

Relative Strength vs SPY

$RS_{20d} = (Price / Price[-20]) - (SPY / SPY[-20]) \quad \leftarrow \text{stock 20d ret minus SPY 20d ret}$
 $RS_{5d} = (Price / Price[-5]) - (SPY / SPY[-5]) \quad \leftarrow \text{5-day version}$
 Positive $\rightarrow$ outperforming the market over the period

Monthly Pivot Points

From prior month's High (H), Low (L), Close (C):

$$PP = (H + L + C) / 3 \quad \leftarrow \text{Pivot Point}$$

$$R1 = 2 \times PP - L$$

$$S1 = 2 \times PP - H$$

$$R2 = PP + (H - L)$$

$$S2 = PP - (H - L)$$

$$R3 = H + 2 \times (PP - L)$$

$$S3 = L - 2 \times (H - PP)$$

Weinstein Stage Analysis

Based on price position relative to 150-day (30-week) moving average:

Stage 1 – Base: Below MA, MA flattening (sideways consolidation)

Stage 2 – Uptrend: Above MA, MA rising, making higher highs and higher lows

Stage 3 – Topping: Above MA but MA starting to flatten/roll over

Stage 4 – Downtrend: Below MA and MA declining

Best entries: Stage 1 → Stage 2 breakout

Avoid longs: Stage 3, Stage 4

Coiled Spring Score (0–100)

Rewards (adds to score):

- + Tight Bollinger Band width (price compression)
- + Drying volume (sellers exhausted)
- + Price near 52-week high
- + Stage 2 classification
- + Positive RS vs SPY

Penalises (subtracts):

- Wide daily range (expanding volatility)
- High volume without price progress
- Far from 52-week high

A high Coiled Spring score means: volatility compressed, momentum building, positioned for a potential explosive move.

Multi-Timeframe Alignment Score

Weekly signals (weight 40%):

- Price above 10-week MA
- 10-week MA slope positive
- Weekly RSI(9) > 50

Daily signals (weight 35%):

- Price above 50-day MA
- Price above 200-day MA
- RS vs SPY 20d > 0

Short-term signals (weight 25%):

- Price above 5-day MA
- 5-day return positive
- Volume confirming

Score = (weekly_hits/3)×0.40 + (daily_hits/3)×0.35 + (st_hits/3)×0.25

Final score scaled 0 → 100

6 · Portfolio & Risk Formulas

Portfolio Return & Variance

$E[R_p] = \sum w_i \times \mu_i$ ← weighted average of expected returns

$\sigma^2_p = \sum_i \sum_j w_i \times w_j \times \text{Cov}(R_i, R_j)$ ← full covariance matrix

$\text{Sharpe} = (E[R_p] - R_f) / \sigma_p$

Backtesting Performance Metrics

$\text{CAGR} = (\text{Terminal_Value} / \text{Initial_Value})^{(1 / \text{Years})} - 1$

$\text{Sharpe} = (\text{Mean_Daily_Return} - R_f) / \text{Daily_Volatility} \times \sqrt{252}$
($R_f = 0$ in most contexts)

$\text{Sortino} = (\text{Mean_Daily_Return} - R_f) / \text{Downside_Volatility} \times \sqrt{252}$
where $\text{Downside_Vol} = \text{std}(\text{only negative returns})$

$\text{Calmar} = \text{CAGR} / |\text{Max_Drawdown}|$

$\text{Max DD}[t] = \text{Portfolio}[t] / \max(\text{Portfolio}[0..t]) - 1$
 $\text{Max DD} = \min(\text{Max DD over all } t)$

$\text{Information Ratio} = \text{mean}(\text{Portfolio Return} - \text{SPY Return}) / \text{std}(\text{Portfolio Return} - \text{SPY Return}) \times \sqrt{252}$

Fama-French 3-Factor Regression

$\text{Excess_Return}[t] = \alpha + \beta_{\text{Mkt}} \times (\text{Mkt-RF})[t] + \beta_{\text{SMB}} \times \text{SMB}[t] + \beta_{\text{HML}} \times \text{HML}[t] + \varepsilon[t]$

Where:

$\text{Mkt-RF} = \text{Market excess return (SPY - risk-free rate)}$

SMB = Small Minus Big: small-cap premium
 HML = High Minus Low: value premium (high Book/Market)

Outputs:

α = Annualised: intercept $\times$ 252 $\rightarrow$ unexplained alpha (skill)
 β_{Mkt} = Market beta (1.0 = same as market)
 β_{SMB} = Size tilt (positive = small-cap bias)
 β_{HML} = Value tilt (positive = value bias)
 R^2 = Fraction of return variance explained by the 3 factors
 t-stats = Statistical significance ($|t| > 2$ is roughly $p < 0.05$)

Value at Risk (VaR) & Conditional VaR

Historical VaR(α):

Sort daily portfolio returns; take the $(1-\alpha)$ percentile
 e.g. 95% VaR = 5th percentile of the return distribution

Parametric VaR(α):

$\text{VaR} = \mu + \sigma \times \Phi^{-1}(1-\alpha)$
 where Φ^{-1} is the inverse normal CDF

Historical CVaR(α):

$\text{CVaR} = \text{mean}(\text{returns} \leq \text{VaR threshold})$
 # Average loss on the worst $(1-\alpha)\%$ of days

Parametric CVaR(α):

$\text{CVaR} = \mu - \sigma \times \phi(\Phi^{-1}(1-\alpha)) / (1-\alpha)$
 where ϕ is the standard normal PDF

Concentration Metrics

HHI (Herfindahl-Hirschman Index) = $\sum w_i^2$

Range: $1/N$ (fully diversified equal-weight) to 1.0 (single position)

Effective N = $1 / \text{HHI}$

Interpretation: equivalent number of equal-weight positions

Example: $\text{HHI}=0.2 \rightarrow \text{Effective N}=5$ (portfolio behaves like 5 equal positions)

7 • Recommended Research Workflow

This is the intended flow from market context to portfolio construction:

#	Step	Tool	Output
1	Morning context — check macro environment, sector rotation, market breadth	Market Overview	Risk-on / risk-off decision; which sectors to favour
2	Universe screening — find stocks with the strongest multi-factor composite scores	Universe Screener (S&P 500 or Theme)	Top 20–50 candidate stocks by Composite Z-score
3	Factor validation — verify the factors you're relying on actually work in the current regime	Factor Explorer	Mean IC > 0.03 and ICIR > 0.5 gives confidence in the signal
4	Technical timing — from your screener shortlist, find stocks at actionable setups	Technical Signals Screener (custom list)	Stocks in Stage 2, above 50/200 MA, near pivot with volume confirmation
5	Portfolio construction — allocate optimally across your final list	Portfolio Optimizer	HRP or Equal Weight allocation (more robust than Max Sharpe out-of-sample)
6	Risk review — check factor loadings, concentration, and tail risk before going live	Risk Dashboard	Alpha > 0, $R^2 < 0.9$ (not just pure beta), CVaR within acceptable limits
7	Backtest validation — verify the strategy has historically worked	Strategy Backtester	Positive Sharpe, positive alpha, IR > 0.5 across multiple market cycles

Quick daily use (3 minutes)

Market Overview → breadth check → Screener sorted by Composite → flag anything at a pivot on the Technical Screener. The full research workflow above is for new position initiation; daily monitoring is just steps 1 + 2 + 4.

8 • Glossary

Term	Definition
Alpha (α)	Return in excess of what can be explained by factor exposures (e.g. market beta). Positive alpha = genuine outperformance.
ATR	Average True Range — measures volatility as the average daily price range, accounting for gaps.
Beta	Sensitivity of a stock's returns to the market (SPY). Beta = 1.5 means the stock moves 1.5% for every 1% market move.
Bollinger Bands	Dynamic price channel: 20-day SMA $\pm$ 2 standard deviations. Measures whether price is extended above/below recent average.
CAGR	Compound Annual Growth Rate — the annual rate that would produce the same cumulative return.
Calmar Ratio	CAGR divided by maximum drawdown. Higher is better — measures return earned per unit of peak loss taken.
CVaR / Expected Shortfall	Average loss on the days that exceed the VaR threshold — the true tail risk metric.
DuckDB	In-process analytical SQL database used to cache all price and fundamental data locally for fast repeated access.
Efficient Frontier	The set of portfolios that offer the maximum return for a given level of risk (or minimum risk for a given return).
Fama-French 3-Factor	Academic model by Fama & French decomposing stock returns into market, size (SMB), and value (HML) factor exposures.
HHI	Herfindahl-Hirschman Index — sum of squared weights. Measures portfolio concentration (0 = fully diversified, 1 = single position).
HML	High Minus Low — the value factor: return of high Book/Market stocks minus low Book/Market stocks.
HRP	Hierarchical Risk Parity — portfolio optimisation that clusters correlated assets and equalises risk contribution at each level of the tree.
IC (Information Coefficient)	Spearman rank correlation between factor scores and subsequent returns. Measures how predictive a factor is.

Term	Definition
ICIR	IC Information Ratio = Mean IC / Std(IC). Measures the consistency (signal-to-noise) of a factor's predictive power.
MACD	Moving Average Convergence/Divergence — momentum indicator using the spread between two exponential moving averages.
Max Drawdown	The largest peak-to-trough decline in portfolio value. The key risk metric for measuring how bad a strategy's worst period was.
RSI	Relative Strength Index — oscillator (0–100) measuring the speed and magnitude of recent price changes. Used to identify overbought/oversold conditions.
Sharpe Ratio	Excess return per unit of total volatility. A Sharpe above 1.0 is considered good; above 2.0 is excellent.
SMB	Small Minus Big — the size factor: return of small-cap stocks minus large-cap stocks.
Sortino Ratio	Like Sharpe, but only penalises downside volatility (negative returns), not upside. Better for skewed distributions.
Stage Analysis	Weinstein's 4-stage model (Base / Uptrend / Top / Downtrend) based on price position relative to 30-week moving average.
VaR	Value at Risk — the maximum expected loss over a period at a given confidence level (e.g. 95%). Does not capture losses beyond this threshold.
Winsorizing	Clipping extreme factor values at ± 3 standard deviations to prevent outliers from distorting the cross-sectional rankings.
yfinance	Open-source Python library that downloads market data from Yahoo Finance. Used as the primary data source.
Z-score	$(\text{Value} - \text{Mean}) / \text{Standard Deviation}$. Standardises a number to how many standard deviations it is above or below the average of its group.